

Knowledge Distillation: Exploring Standard, Reversed, and Self-Distillation Techniques

Reversed Distillation

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Abstract

Convolutional neural networks achieve state-of-the-art results in image classification but often exceed the resource limits of mobile and embedded devices. Knowledge Distillation (KD) compresses these large models by transferring soft targets from one or more “teacher” networks into a smaller “student” network. In this work, we reproduce and extend four KD paradigms—Reversed KD , Multi-Teacher KD , Combining quantization and distillation, and Teacher-Free Self-Distillation — and introduce preliminary adversarial training for enhanced robustness. These findings align with broad knowledge of KD’s utility across vision, natural language processing, and speech recognition applications.

Distillation Basics

The network architectures and corresponding MNIST accuracies, with and without self-distillation, are summarized in Table 1 [1] and Table 2. The results clearly demonstrate that soft labels produced by the larger model contribute to the improved accuracy of the student model. Moreover, when using ResNet18 as the teacher model, the performance gain observed in the student model is less significant compared to when both teacher and student models were MLPs.

Table 1: Architecture of teacher and student models

Model	Architecture	FLOPs	#Prameter
Teacher Model 1:MLP	(fc1): Linear(in_features=784,out_features=1200,)	2.4 MMac	2.4 M
	(fc2):Linear(in_features=1200,out_features=1200)		
	(fc3): Linear(in_features=1200,out_features=10)		
Student Model 1:MLP	(fc1): Linear(in_features=784,out_features=400)	318.41KMac	318.01 k
	(fc2): Linear(in_features=400,out_features=10)		
Teacher Model 2: ResNet18	ResNet18 backbone for MNIST (conv1, 4 residual layers + FC)	457.65 MMac	11.17 M

Table 2: Distillation vs. Baseline Student Performance

Model	Teacher 1	Teacher 2	Student baseline	Distillation 1	Distillation 2
Test accuracy	98.96	99.38	98.16	98.87	98.33

Figure 1: Effect of Temperature on Knowledge Distillation Accuracy

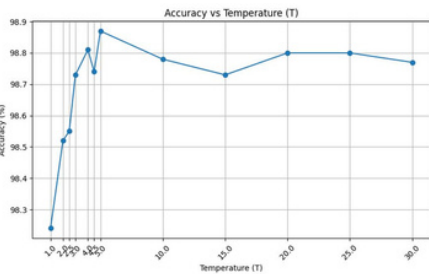


Figure 1 shows the relationship between the distillation temperature T and the student model's test accuracy. We observe that the accuracy initially increases with higher temperature, peaking at T=5 with an accuracy of 98.87%, which is a +0.71% improvement over the baseline.

Table 3: Performance on Omitted Digit

Training Method	Training Set	Overall Test Acc	Acc on Digit
Baseline (Cross-Entropy)	MNIST without '3'	88.24	0.00
Distillation (T=10)	MNIST without '3'	98.56	95.05
Baseline (Cross-Entropy)	MNIST without '4'	88.61%	0.00
Distillation (T=10)	MNIST without '4'	98.64%	95.62

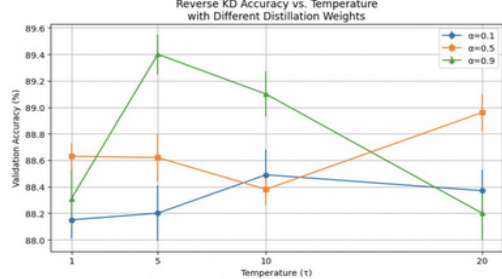
Table 3 illustrates the effectiveness of knowledge distillation when a specific digit ('3' or '4') is omitted from the student's training set. The baseline models trained using cross-entropy fail to recognize the omitted digit, achieving 0% accuracy on it. In contrast, student models trained via distillation (with T=10) achieve over 95% accuracy on the unseen digits, while also improving overall test accuracy by over 10%.

Table 4: Reverse KD: Baseline vs. Distilled (Validation Accuracy, %)

Model	ResNet-34	ResNet-50	Re-KD
Mean ± Std	89.02 ± 0.23	88.32 ± 0.25	89.16 ± 0.23

Re-KD is a network architecture in which a student teaches a teacher [2]. Over three seeds, the ResNet-34 teacher averaged 89.02 ± 0.23 % and the ResNet-50 student’s standalone baseline was 88.32 ± 0.25 %. After Re-KD, the student rose to 89.16 ± 0.23 %, consistently beating its own baseline and even edging past its smaller teacher, showing that “reverse” distillation can successfully transfer a compact model’s dark knowledge to improve a larger one.

Figure 2: Hyperparameter Sweep (Reverse KD)



This sweep ($\tau \in \{1, 5, 10, 20\}$, $\alpha \in \{0.1, 0.5, 0.9\}$) measures Re-KD accuracy across temperatures τ and distillation weights α . Peak performance (89.40 ± 0.15 %) occurs at $\tau = 5$, $\alpha = 0.9$, while extremes of τ or low α underperform. Hence, moderate softening plus heavy soft-target weighting best transfers dark knowledge.

Self-Distillation (Teacher-free)

Self-Distillation (SD) explores how a model can improve itself by learning from its own internal representations or predictions, eliminating the need for a separate teacher. SD Architecture: Auxiliary classification heads were added after ResNet blocks layer2 and layer3. The final classification layer acts as the teacher during training. [2]

weighted combination Loss Function:

$$L_{total} = L_{final_CE} + W_{aux} \sum L_{aux_CE} + W_{kd} \sum L_{KD(T)}$$

Method	Baseline (ResNet-50)	Self-distillation (T=3.0, W_aux=0.4, W_kd=0.6)	Improvement vs Baseline
Clean Test Accuracy	87.68%	87.94%	+0.26 pt

Table 5: comparison between baseline and Self-KD ResNet 50

The table compares the clean test accuracy of a Baseline ResNet-50 model (87.68%) against an initial Self-Distillation setup (using T=3.0, W_aux=0.4, W_kd=0.6), which achieved 87.94% accuracy.

Quantization on distillation

Table 6: Quantized Distilled Student Performance

Bit Depth	Quantized Test Acc (%)	Accuracy Drop vs FP32 (%)
FP32	98.54	-
INT8	98.51	-0.03
INT4	98.35	-0.19

We observe that quantizing the distilled model from FP32 to INT8 results in a minimal drop in accuracy of only 0.03%, indicating that 8-bit quantization preserves almost all of the performance benefits of full-precision training. Even with more aggressive quantization to 4-bit (INT4), the model only experiences a 0.19% decrease in accuracy. These results are consistent with the findings in Polino [4].

Multi-Teacher Knowledge Distillation

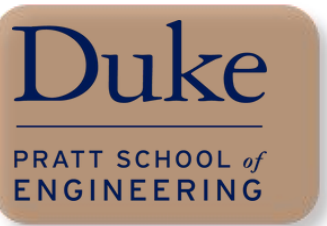
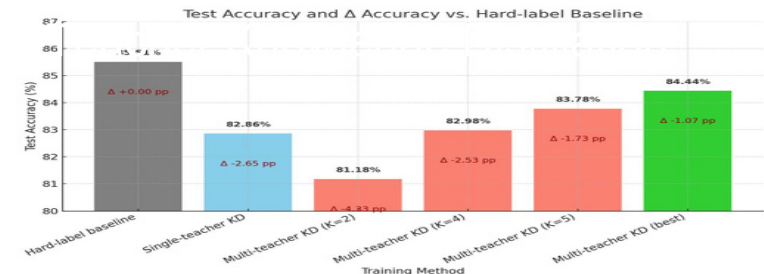


Figure4: Comparison between three training schemes on CIFAR-10

Multi-Teacher KD averages soft targets from multiple pre-trained teachers to train a student, smoothing individual biases. On CIFAR-10, a hard-label baseline achieves 85.51%. Single-teacher KD (K=1) yields 82.86%, while multi-teacher KD (K=2,4,5) recovers to 83.78% (K=5) and, after tuning, 84.44%. Though none surpasses the baseline, multiple teachers reduce accuracy loss (~1.6 pp gain from K=1 to 5), suggesting that ensemble soft targets better regularize students. Per- K tuning may further close or reverse the gap.

Self-KD Hyperparameter tuning

Experiment Description	Aux Temp (T)	CE Weight (β)	KD Weight (α)	Rationale/Hypothesis	Initial Run Acc (%)	Final Test Acc (%)	Improvement vs Baseline (%)
Initial SD Run	3	0.4	0.6	(Completed)	87.94	87.94	+0.26%
Tuning Exp 1	2	0.4	0.6	(Lower T)	88.23	88.23	+0.55%
Tuning Exp 2	5	0.4	0.6	(Higher T)	87.31	87.31	-0.37%
Tuning Exp 3: Increase KD Weight	3	0.3	0.7	Test sharper distribution	87.39	87.39	-0.29%
Tuning Exp 4: Balance Weights	3	0.5	0.5	Equal weight for Aux CE and KD	87.24	87.24	-0.44%
Tuning Exp 5: Sharp T	1	0.4	0.6	Test standard softmax (T=1.0)	88.29	88.29	+0.61%

Table 7: Hyperparameter Tuning for Self-KD

Goal: To find the optimal Temperature (T), Auxiliary CE Weight (Waux), and KD Weight (Wkd) for the Self-Distillation process. **Experiments:** Several combinations were tested, primarily varying T while keeping weights at (0.4, 0.6), and varying weights while keeping T. **Outcome:** The best performance, **88.29%** test accuracy, was achieved with **T=1.0, Waux=0.4, Wkd=0.6**. This configuration yielded the highest improvement over the baseline (+0.61%). **Trend:** Lower temperatures (T=1.0 and T=2.0) performed better than higher temperatures (T=3.0, T=5.0) or different loss weight balances in this series of experiments.

Robustness Test

Model	Clean Test Acc (%)	Avg. Acc on CIFAR-10-C (%)	Avg. Diff. on C (vs Baseline) (%)	Relative Accuracy (%)
Baseline	87.68%	73.37%	-	83.68%
Best Self-KD	88.29%	74.23%	+0.86 points	84.08%

Table 8: Robust Test for the best-tuned Self KD model

We conduct the robustness evaluation comparing the Baseline ResNet-50 against the best-tuned Self-Distilled (SD) model on the CIFAR-10-C dataset. The result is shown in the table (74.23% vs 73.37%), which shows Self-KD’s advantage of an improvement of +0.86 percentage points).

Conclusion

Knowledge Distillation (KD) enables compact models to inherit performance from larger networks. Our study validates several KD paradigms including Reversed KD, Multi-Teacher KD, and Self-Distillation, each offering varying improvements. Furthermore, quantizing distilled models to INT8 or even INT4 maintains high accuracy, demonstrating that KD pairs well with model compression. We also explore adversarial robustness and observe that KD-trained models exhibit stronger generalization under perturbations. Overall, KD proves effective across diverse configurations, delivering lightweight yet accurate networks suitable for real-world deployment on resource-constrained devices.

References

[1] Hinton, Geoffrey, Oriol Vinyals, and Jeff Dean. "Distilling the knowledge in a neural network." arXiv preprint arXiv:1503.02531 (2015).
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[5] Yang, C., Xie, N., Wang, W., & Wang, J. (2019). "Snapshot ensemble distillation: Teacher-student optimization in one generation." Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR).